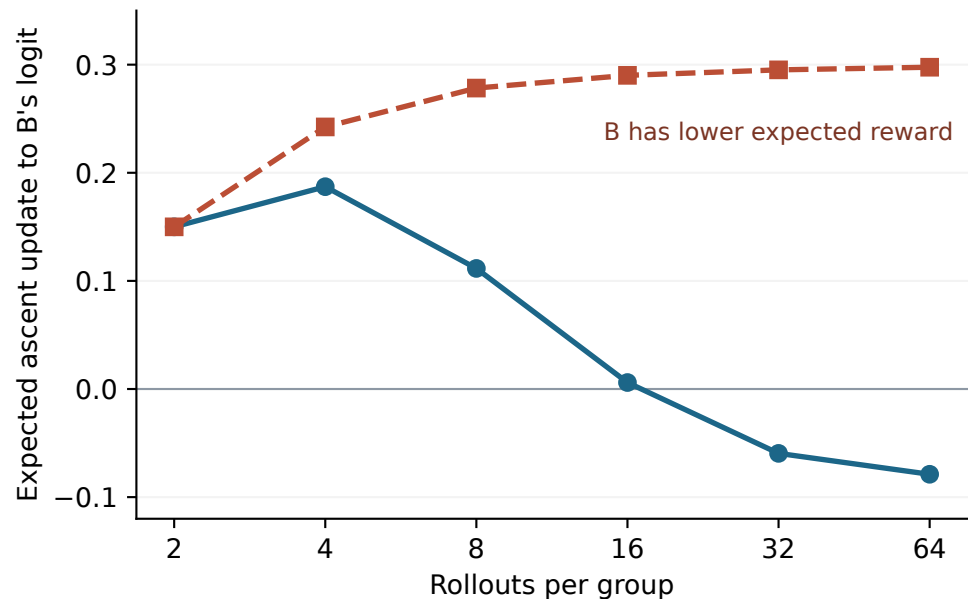


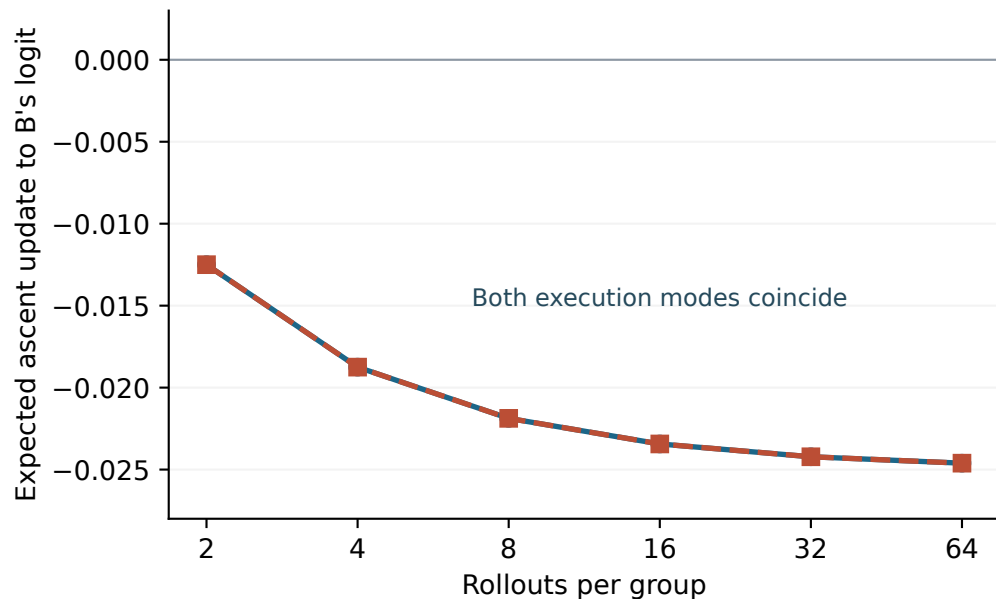
Sharing stochastic tool results can reverse the update

Constructed bandit: A always returns 0.9; B succeeds with probability 0.8. Initial $P(B) = 0.5$.

Group-normalized update



Centered-only control



—●— Independent tool outcomes —■— One shared outcome per group

Exact finite sums in floating arithmetic. Mean reward is 0.85 in both modes. No language model was trained.